MIL-S- 46319(Ord)
31 July 1961

SUPERSEDING
FA(FC)-PD-14
14 June 1957

MILITARY SPECIFICATION

SIGHT: M39

1. SCOPE

1.1 This specification covers one type of instrument known as Sight M39, used as a sighting and aiming device for the 3.5 inch Rocket Launchers, M20 Series.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-O-13830	Optical Components for Fire Control Instruments: General Specification Governing the Manufacture, Assembly and Inspection of
MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-13968	Sight M39, Packaging of
MIL-I-45208	Inspection Requirements, General Specification for
MIL-I-45607	Inspection Equipment, Supply and Maintenance for Ordnance

FSC 1240

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130	Identification Marking of U.S. Military Property

DRAWINGS

Ordnance Corps

F7162978	Sight: M39
F7659264	Inspection Fixture

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Production model.- At the beginning of production one sight, representative of production, shall be made available to the Government to determine if the sight meets all requirements of the specification.

3.2 Materials.- Materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

3.3 Construction.- The sight shall be manufactured in accordance with Drawing F7162978 "Sight: M39" and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired optical and mechanical characteristics, performance and permanence to insure inherent quality and performance capabilities prior to final acceptance by the Government.

3.4 General specifications.- The contractor shall be responsible for adherence to and compliance with the following requirements of Specifications MIL-O-13830 and MIL-F-13926.

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3.4.1 MIL-O-13830

(a) Cleanliness
(b) Illuminated reticle

3.4.2 MIL-F-13926

(a) Order of precedence
(b) Protective finished surfaces
(c) Special accuracy
(d) Finish painting and protective
(e) Workmanship

3.5 Environmental.

3.5.1 Storage temperatures.- The sight shall be exposed and thermally stabilized at ambient temperatures of minus 80 degrees and plus 160 degrees Fahrenheit. Upon return to standard ambient temperature (plus 60 degrees to 90 degrees Fahrenheit) the sight shall show no indication of physical failure and shall meet all requirements of this specification.

3.5.2 Operating temperatures.- The sight shall meet the requirements of 3.16 while exposed and thermally stabilized at ambient temperatures of plus 150 degrees and minus 40 degrees Fahrenheit. Upon return to standard ambient temperature (plus 60 to 90 degrees Fahrenheit) the sight shall meet the requirements of 3.5.3.

3.5.3 Vibration.- With the sight mounted in its normal operating position it shall be vibrated in a vertical plane at a constant frequency of thirty cycles per second with an amplitude of 1/16 inch (1/8 inch total excursion) for a period of five minutes plus or minus fifteen seconds. Subsequent to vibration there shall be no evidence of damage or physical failure and the sight shall meet the requirements of 3.5.5 to 3.18 inclusive.

3.5.4 Humidity.- The sight shall be exposed to one complete humidity cycle as shown on Figure I. Immediately following the completion of the cycle the sight shall be inspected and the optical surfaces shall show no evidence of moisture.

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3.5.5 Cleanliness.- Cleanliness shall be in accordance with the cleanliness requirements of Specification MIL-O-13830 and there shall be no more than two particles of dirt visible on the surface of the reticle and the area of each particle shall not be in excess of the area equivalent to a size five dig. There shall be no dirt on other surfaces which shall interfere with observation or adversely affect image quality when viewing through the sight.

3.5.6 Sealing.- The sight shall show no evidence of leakage when subjected to an internal pressure of 5 psi (gage pressure) using dry nitrogen over a two minute period, after which the sight shall be bled to zero psi (gage pressure).

3.6 Collimation.- The requirements of 3.6.1 to 3.9 inclusive shall be met with the sight assembled as shown on Drawing F7162978 (sheet 2) in its normal operating position under the following conditions:

(a) Sight secured by means of the threaded stud on Post D8567788 with the stud axis horizontal.
(b) Bracket Assembly C8567787 secured with its locating surface in a vertical plane.
(c) The center of the center mounting hole in coincidence with a vertical line passing through the axis of the threaded stud in (a) above.
(d) The centers of the two locating holes on each side of the center locating hole in a horizontal plane.
(e) Plunger in zero detent.

3.6.1 Elevation boresight.- The elevation boresight adjusting screw shall be actuated to bring the line of sight through the boresight point on the reticle in coincidence with a horizontal target point. From this position further actuation of the elevation boresight adjusting screw shall move the line of sight at least 25 mils above and 25 mils below the horizontal target point.

3.6.2 Azimuth boresight.- The azimuth boresight adjusting screw shall be actuated to bring the line of sight through the boresight point on the reticle, in coincidence with a vertical target point that is 3.1425 inches to the left of and parallel to the vertical mounting surface. From this position further actuation of the azimuth boresight adjusting screw shall move the line of sight at least 25 mils to the left and 25 mils to the right of the aiming point on the vertical target line.

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3.7 Parallelism of reticle and image.- With the sight positioned as in 3.6 the zero range line (across entire pattern) shall be parallel to the image of a horizontal line within 2 mils (block) at one extremity of the reticle line when the other extremity of the reticle line is in coincidence with a horizontal target line.

3.8 Plumb travel.- With the sight positioned as in 3.6 and with the line of sight through the zero aiming point in coincidence with a plumb line, the line of sight shall not deviate from the plumb line by more than 1.5 mils (total excursion) when the sight is depressed to the "9" detent.

3.9 Pivoting.- Pivoting the sight into its closed position and returning to the firing position shall not cause the line of sight through the boresight mark to deviate from the initial position by more than 1.0 mil in azimuth and 0.5 mil in elevation.

3.10 Definition.- Resolving power on the optical axis shall be 20 seconds of arc or better.

3.11 Parallax.- With a target at 250 yards plus or minus 25 yards, parallax on the optical axis shall not exceed 0.25 mil.

3.12 Reticle accuracy.- The angular separation between the zero and 500 yard range mark shall be as shown on the reticle drawing within plus or minus 2 mils.

3.13 Image tilt.- The image of a vertical target shall be vertical within one degree (one degree is equivalent to the angle subtended by a 7 mils (block) at the edge of the field when measured over the full field of view).

3.14 Eyepiece focus.- The eyepiece focus shall be fixed between the limits of minus 0.75 and minus 1.0 diopter.

3.15 Field of view.- There shall be a circular field of view having a minimum of 22 degrees with no indication of cut off.

3.16 Operability.- The sight shall be mechanically operable by actuation of the sight over its elevation range from "0" to "9" detent and return, and shall pivot from the closed to firing position at the standard ambient temperature and extreme temperatures of 3.5.2.

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3.17 Illumination.- When illuminated by Instrument Light 8246214 or equivalent, the reticle marking shall appear clearly defined when observed in a darkened area.

3.18 Identification marking.- Identification marking unless otherwise specified shall be in accordance with Standard MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

4.1 General provisions.- The quality assurance provisions of referenced documents to the extent specified herein, together with provisions of this section, are applicable to inspection performed by the contractor and for Government verification.

4.1.1 The contractor shall provide an inspection system in accordance with Specification MIL-I-45208.

4.2 Production model.- Unless otherwise specified in the contract, one sight shall be selected by the Government inspector as soon as practicable after the start of production, and be inspected by the contractor under the surveillance of the Government. This inspection shall be conducted at the contractractor's facility or Government approved laboratory, using test equipment authorized for use on the contract. The model shall be inspected for all requirements of this specification by utilizing examination and testing methods prescribed herein.

4.2.1 Production model failure.- Should the production model fail to pass the prescribed tests, the contractor shall cease those operations which are causing the rejection and shall make the necessary corrections. When corrections in the process have been made, a new production model shall be examined and tested by the contractor under the surveillance of the Government. Acceptance of production lots will not commence until an acceptable production model is submitted and corrective actions have been taken to eliminate deficiencies.

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4.3 Lot formation.- Inspection lot sizes and lot formation shall be in accordance with Standard MIL-STD-105.

4.3.1 Defective units and lots shall be processed as specified in Standard MIL-STD-105.

4.4 Material.- Control of material shall be in accordance with test data requirements of Specification MIL-I-45208.

4.5 Inspection equipment.- Inspection equipment shall conform to drawings listed in this specification, except as provided for under the alternative inspection procedures and inspection equipment requirement of MIL-I-45208. Provisions of MIL-I-45607 shall apply to test equipment provided by the Government or provided by the contractor and utilized in the Government approved inspection system. The Government reserves the right to use all approved test equipment, both Government furnished and contractor acquired, for verification inspection.

4.5.1 Accuracy requirements.- Inspection equipment which incorporates features for measuring values or sizes established by product requirements shall be capable of measuring those values within an accuracy of ten percent of the specified tolerance, except as otherwise specified. In cases where the ten percent accuracy imposes impractical or very difficult design and fabrication problems, or results in a test equipment cost factor which is exorbitant in relation to the cost and application of the product to be inspected, the Government shall be expeditiously advised in order that necessary appropriate action can be taken.

4.6 Tests.

4.6.1 Classification of defects.- The classification of defects in Tables I and II shall constitute the minimum inspection requirements to be performed prior to acceptance or rejection by lots. The Government reserves the right to inspect for any or all applicable requirements and to reject individual non-conforming items.

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4.6.1.1 Disposition of rejected lots.- Rejected lots shall be screened for all defective characteristics. After corrections have been made and the Government inspector informed of the corrective action taken, the lots shall be resubmitted for acceptance as outlined in Standard MIL-STD-105.

TABLE I - CLASSIFICATION OF DEFECTS

Use Inspection Level L8, and Sampling Plan Table IV-A of Standard MIL-STD-105

CRITICAL: NONE DEFINED

MAJOR: AQL 1.5%		Req't Par.	QA. Par.	Test Method
101.	Storage temperature	3.5.1	4.7.1.1	Device, Special Testing (DST)
102.	Operating temperature	3.5.2	4.7.1.2	DST
103.	Vibration	3.5.3	4.7.1.3	Standard Measuring Equipment (SME)
104.	Humidity	3.5.4	4.7.1.4	DST
105.	Cleanliness	3.5.5	4.7.1.5	Visual
106.	Sealing	3.5.6	4.7.1.6	SME

TABLE II - CLASSIFICATION OF DEFECTS

Use Inspection Level II, and Sampling Plan Table IV-A of Standard MIL-STD-105

CRITICAL: NONE DEFINED

MAJOR: AQL 1.5%		Req't Par.	QA. Par.	Test Method
101.	Collimation	3.6	4.7.2	Device, Special Testing (DST)
102.	Elevation boresight	3.6.1	4.7.3	DST
103.	Azimuth boresight	3.6.2	4.7.4	DST
104.	Parallelism of reticle and image	3.7	4.7.2	DST
105.	Plumb travel	3.8	4.7.2	DST
106.	Pivoting	3.9	4.7.2	DST

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TABLE II - CLASSIFICATION OF DEFECTS (Cont'd)

MAJOR: AQL 1.5%		Req't Par.	QA. Par.	Test Method
107.	Definition	3.10	4.7.2	DST
108.	Parallax	3.11	4.7.5	Visual
109.	Reticle accuracy	3.12	4.7.2	DST
110.	Image tilt	3.13	4.7.2	DST
111.	Eyepiece focus	3.14	4.7.6	Standard Measuring Equipment (SME)
112.	Field of view	3.15	4.7.2	DST
113.	Operability	3.16	4.7.7	Manual
114.	Illumination	3.17	4.7.8	Visual
115.	Identification marking	3.18	4.7.9	Visual
116.	Post D-8567788 (NS-1/2-40)	3.3	4.7.10	Gage

4.7 Test methods and procedures.

4.7.1 Environmental.

4.7.1.1 Storage temperature.- The sight shall be placed in a test chamber capable of the temperature requirements of 3.5.1. The test shall be in accordance with Procedure II of Specification MIL-F-13926. After exposure, the sight shall be visually inspected for damage or physical failure, then shall meet the requirements of 3.5.1.

4.7.1.2 Operating temperatures.- The sight shall be placed in its normal operating position (see Dwg. F7162978) utilizing a test chamber capable of maintaining temperatures specified in 3.5.2. The test shall be conducted in accordance with Procedure I of Specification MIL-F-13926 and shall meet the requirements indicated in 3.5.2.

4.7.1.3 Vibration.- The sight shall be mounted in its normal operating position (see Dwg. F7162978) by means of a vibrating machine capable of vibrating the sight in a vertical plane with an amplitude, duration and frequency required in 3.5.3. At the conclusion of this test, the sight shall be visually inspected for damage or physical failure, then shall meet the requirements of 3.5.3.

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4.7.1.4 Humidity.- The sight shall be placed in a test chamber so that it is no closer than six inches from the chamber wall and is not subject to radiant heat. Distilled or demineralized water having a P_H value between 6.5 and 7.5 at 77 degrees Fahrenheit shall be used to obtain the specified humidity. The sight shall be exposed to the humidity condension cycle as indicated on Figure 1. Upon completion of the cycle, the sight shall show no evidence of damage or moisture penetration.

4.7.1.5 Cleanliness.- Cleanliness is a visual test made in accordance with Specification MIL-O-13830 for requirements of 3.5.5.

4.7.1.6 Sealing.- Sealing test is accomplished by means of commercial equipment, which shall include a sensitive pressure gage, capable of testing for requirements of 3.5.6. The equipment is attached to the sight by means of the 8 x 32 threaded hole in Cover D-8212973 provided for this purpose. The pressure of dry nitrogen indicated in 3.5.6 is applied and then the source cut off. Evidence of leakage can be determined by observing the gage for drop of pressure and time interval as specified.

4.7.2 Collimation.- Collimation test is made by means of a special testing device depicted on Drawing F-7659264. This drawing has set-up instructions for the testing device and operating instructions for making the following tests:

Collimation
Parallelism of reticle and image
Plumb travel
Pivoting
Definition
Reticle accuracy
Image tilt
Field of view

4.7.3 Elevation boresight.- Elevation boresight adjusting screw is checked for actuation of at least 25 mils above and below the boresight point by means of the same set-up as used for the collimation test. The 25 mil marks are indicated on the wall target Drawing F-8272082.

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4.7.4 Azimuth boresight.- Azimuth boresight adjusting screw is checked for actuation of at least 25 mils to the left and right of the boresight point of the reticle by means of the same set-up as used for the collimation test. The 25 mil marks are indicated on the wall target Drawing F-8272082, and the 3.1425 inch dimension to the left of the mounting surface is compensated by the location of the wall target.

4.7.5 Parallax.- This test shall be accomplished by sighting the sight to a target at the distance specified in 3.11. Parallax is recognized as any apparent movement of the reticle line in relation to the target image when the observer's head is moved slightly from side to side or up and down. Parallax is checked at the optical axis (center of reticle pattern) and shall not exceed 1/2 the width of the reticle line for requirement of 3.11.

4.7.6 Eyepiece focus.- Eyepiece focus test is made with the aid of a 3-power dioptometer adjusted for the observer's eye accommodation. Locate the sight with the objective facing an illuminated white background so that the reticle pattern is clearly visible, and set the diopter scale at zero. Place the dioptometer in front of the eyepiece and remove parallax between reticle by movement of the dioptometer scale. The reading of the dioptometer scale will indicate the error which shall be within the limits of 3.14.

4.7.7 Operability.- Operability is a functional test made at standard ambient temperature and extreme temperatures of 3.5.2.1 for the operability of the elevation range and pivot movement. The test shall be made with hand pressure and without the use of tools.

4.7.8 Illumination.- Illumination test is a visual check made within the confines of a dark room for requirements of 3.17.

4.7.9 Identification marking.- Identification marking test is a visual check made in accordance to Standard MIL-STD-130.

4.7.10 Post D-8567788.- The post shall be tested for the 1/2" x 40NS thread by means of a thread gage to insure interchangeability in compliance with the construction requirements of 3.3.

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5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with Specification MIL-P-13968. The level of protection shall be as specified in the procurement documents.

6. NOTES

6.1 Intended use.- The Sight M39 is primarily intended as the sighting and aiming device for direct fire of the 3.5 inch Rocket Launchers of the M20 Series which is a shoulder supported weapon.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title number and date of this specification
(b) Selection of applicable levels of preservation, packaging, and packing.
(c) Applicable stock numbers.
(d) Provisions for production model including date of delivery (see 3.1)

6.3 Design information.- Design information regarding specific equipment not covered or referenced by this specification may be obtained from the procuring activity upon specific request of the contractor.

6.4 Definitions.

6.4.1 Standard measuring equipment (SME).- Standard measuring equipment (SME) is defined as the common hand type measuring devices which are usually stocked by commercial supply houses for ready supply (shelf items), and which are normally used by an inspector to perform dimensional inspection of items under procurement. This category also includes commercial testing equipment such as meters, optical comparators, etc.

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6.4.2 Special testing device (DST).- Special testing devices are defined as special inspection fixtures, instruments, and modified standard measuring equipment.

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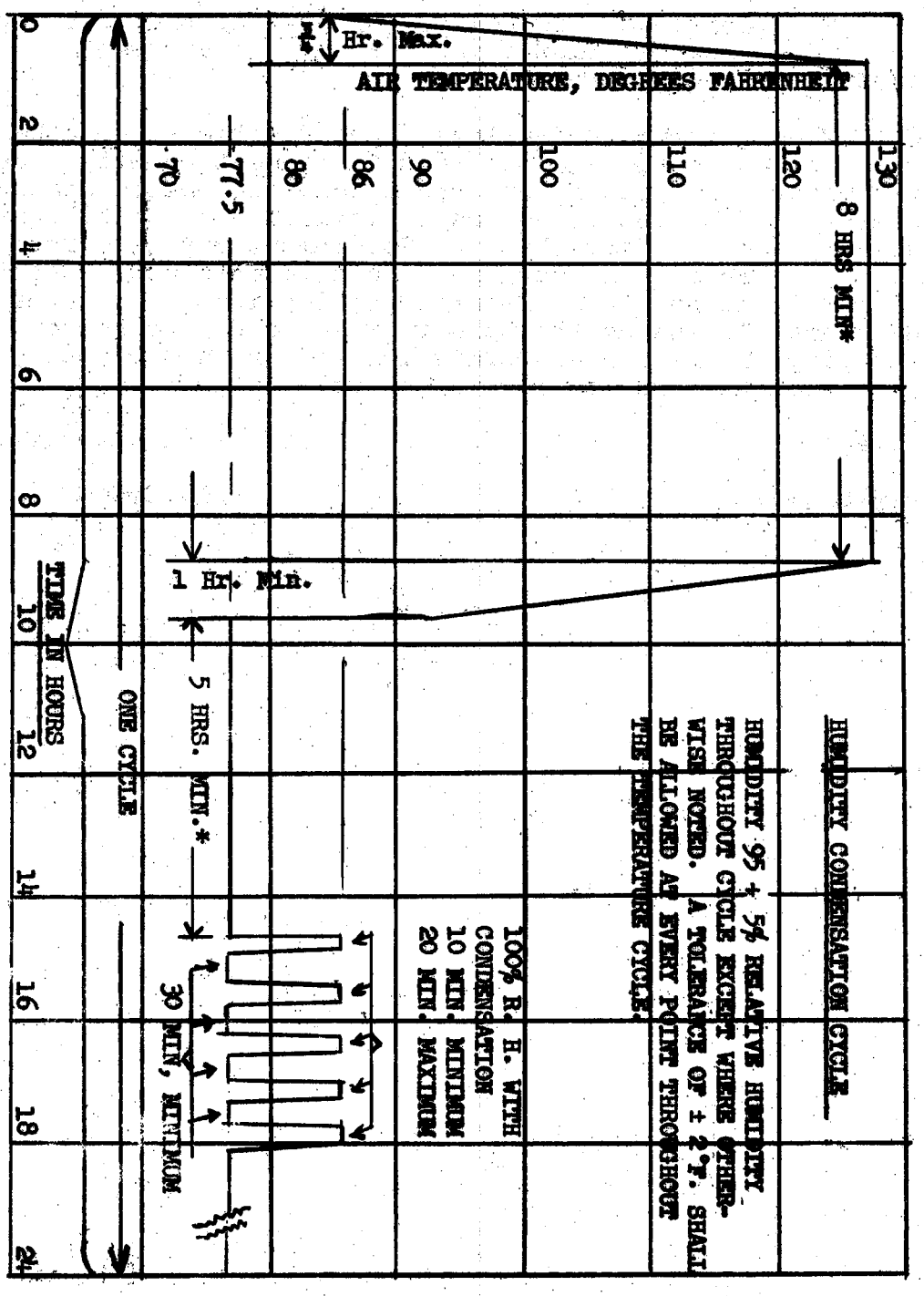
Custodian:

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* OR UNTIL EQUIPMENT STABILIZATION (WHICHEVER IS LONGER)

FIGURE 1

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